

Lab 9.3.8 Report Exploring Nmap

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Part 1: Exploring Nmap

1.1 Understanding Nmap

Question 1: What is Nmap?

Nmap (Network Mapper) is a free and open-source network scanning tool used for network discovery and security auditing. It is designed to rapidly scan large networks and determine which hosts are available, what services they are offering, what operating systems they are running, and what type of firewalls are in use.

Question 2: What is nmap used for?

Nmap is used for:

- Network inventory and asset management
- Discovering open ports on target hosts
- Identifying services and their versions running on those ports
- Detecting operating systems and their versions
- Security auditing and vulnerability assessment
- Network mapping and topology discovery
- Monitoring service uptime

1.2 Nmap Command Switches

Question 3: What is the nmap command used in Example 1?

```
nmap -A -T4 scanme.nmap.org
```

Question 4: What does the switch -A do?

The -A switch enables aggressive scanning options including:

- OS detection (-O)
- Version detection (-sV)

- Script scanning (-sC)
- Traceroute (--traceroute)

This option enables comprehensive reconnaissance of the target.

Question 5: What does the switch -T4 do?

The -T4 switch sets the timing template to "aggressive" mode. It speeds up the scan by using faster timeouts, sending packets more quickly, and reducing delays between probes. T4 is suitable for reliable, fast networks. The timing scale ranges from T0 (paranoid/slowest) to T5 (insane/fastest).

Part 2: Scanning for Open Ports

2.1 Localhost Scan

Command Used:

```
nmap -A -T4 localhost
```

Scan Results:

Port	State	Service	Version
21/tcp	Open	FTP	vsftpd 2.0.8 or later
22/tcp	Open	SSH	OpenSSH 8.2 (protocol 2.0)
23/tcp	Open	Telnet	Openwall GNU/*/Linux telnetd

Question 1: Which ports and services are opened?

Based on the scan results, the following ports are open:

- Port 21/tcp - FTP service
- Port 22/tcp - SSH service
- Port 23/tcp - Telnet service

Question 2: Software providing the services:

- Port 21/tcp: vsftpd 2.0.8 or later (FTP server)
- Port 22/tcp: OpenSSH 8.2 (protocol 2.0)

- Port 23/tcp: Openwall GNU/*/Linux telnetd

Analysis:

The localhost scan revealed three open ports. The FTP service allows anonymous login, which could be a security concern. The presence of Telnet (port 23) is particularly concerning as it transmits data in plaintext, making it vulnerable to interception.

2.2 Local Network Scan

Network Configuration:

ip address

Results:

- IP Address: 127.0.0.1

Question: Which network does your VM belong to?

- The VM belongs to the 127.0.0.1/8

network.

Network Scan Command:

- `nmap -A -T4 127.0.0.1/8`

Scan Duration: Approximately 346.89 seconds

Question 1: How many hosts are up?

According to the scan results, 4 hosts are up on the network.

10.0.2.15 FTP (vsftpd), SSH (OpenSSH 8.2), Telnet

Analysis:

The network scan identified four active hosts. The CyberOps Workstation VM (10.0.2.15) shows the same services as the localhost scan, which is expected. Other hosts on the network include the gateway and potentially other virtual machines or physical devices.

2.3 Remote Server Scan

Target: scanme.nmap.org

Question: What is the purpose of this site?

The purpose of scanme.nmap.org is to provide a legal and authorized target for users to practice and test Nmap scanning. It is maintained by the Nmap project specifically for testing purposes. Users are allowed to scan this server to learn and improve their Nmap skills without violating any laws or ethical guidelines.

Scan Command:

```
nmap -A -T4 scanme.nmap.org
```

Question 1: Which ports and services are opened?

Port	State	Service	Version
22/tcp	Open	SSH	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.8
80/tcp	Open	HTTP	Apache httpd 2.4.7 (Ubuntu)
9929/tcp	Open	nping-echo	Nping echo
31337/tcp	Open	tcpwrapped	-

Question 2: Which ports and services are filtered?

Port	Service	Status
25/tcp	SMTP	Filtered
135/tcp	MSRPC	Filtered
139/tcp	NetBIOS-SSN	Filtered
445/tcp	Microsoft-DS	Filtered
593/tcp	HTTP-RPC-EPMAP	Filtered
4444/tcp	Krb524	Filtered

Question 3: What is the IP address of the server?

The IP address of scanme.nmap.org is: 45.33.32.156

Question 4: What is the operating system?

The operating system is: Linux (Ubuntu)

Analysis:

The remote server scan shows a well-configured security posture. Only necessary services (SSH and HTTP) are open, while potentially dangerous Windows services (135, 139, 445) are filtered by a firewall. The Apache web server displays a message "Go ahead and ScanMe!" confirming this is an authorized scanning target.

Reflection and Analysis

Security Implications

How can Nmap help with network security?

Network administrators and security professionals use Nmap to:

1. **Vulnerability Assessment:** Identify open ports and outdated services that could be exploited
2. **Security Auditing:** Verify that only necessary services are running and accessible
3. **Firewall Testing:** Confirm firewall rules are properly configured and effective
4. **Network Inventory:** Maintain accurate records of all devices and services on the network
5. **Compliance Verification:** Ensure systems meet security standards and policies
6. **Incident Response:** Quickly identify compromised systems or unauthorized services
7. **Penetration Testing:** Simulate attacker reconnaissance to find weaknesses before real attackers do

How can Nmap be used by a threat actor as a nefarious tool?

Malicious actors can abuse Nmap to:

1. **Reconnaissance:** Map target networks to identify potential attack vectors
2. **Vulnerability Discovery:** Find exploitable services and outdated software versions
3. **Target Selection:** Identify the weakest systems to compromise first

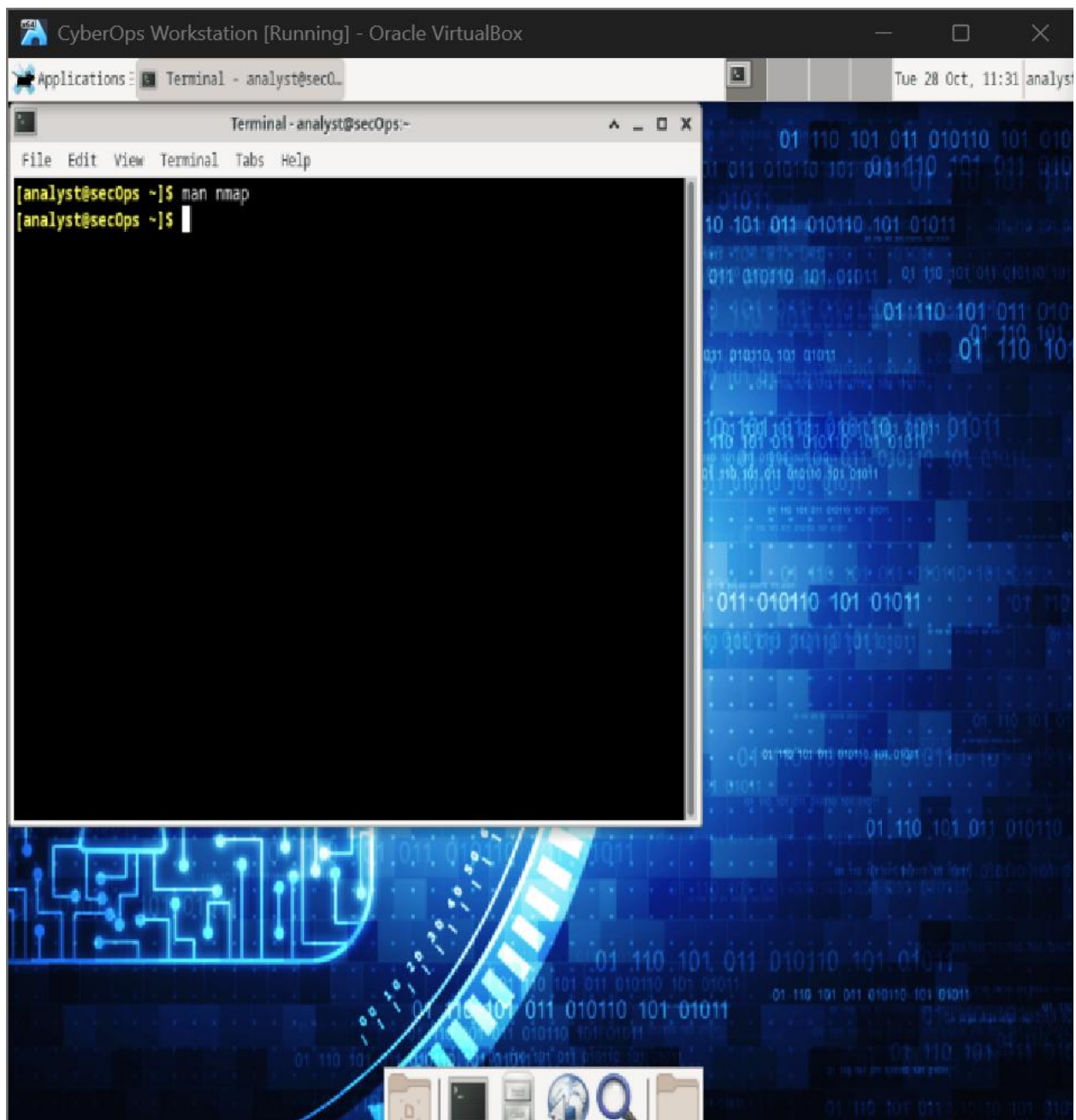
4. **Attack Planning:** Gather intelligence about network topology and security measures
5. **Stealth Scanning:** Use slow scan techniques to avoid detection while gathering information
6. **Service Fingerprinting:** Identify specific software versions to target known vulnerabilities
7. **Network Mapping:** Create detailed maps of target infrastructure for coordinated attacks

Ethical Considerations

Nmap is a dual-use tool - its value depends entirely on the intent and authorization of the user. The same capabilities that help defenders secure networks can be used by attackers to compromise them. This highlights the importance of:

- Always obtaining proper authorization before scanning any network
 - Using security tools responsibly and ethically
 - Implementing defensive monitoring to detect malicious scanning activities
 - Understanding the legal implications of unauthorized network scanning
 - Maintaining strong network security to defend against reconnaissance attempts
-

Screenshots and Evidence



CyberOps Workstation [Running] - Oracle VirtualBox

Applications: Terminal - analyst@sec0... Tue 28 Oct, 11:33 analyst

Terminal - analyst@secOps:~

File Edit View Terminal Tabs Help

```
[analyst@secOps ~]$ man nmap
[analyst@secOps ~]$ nmap -A -T4 Localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2025-10-28 11:32 EDT
Nmap scan report for Localhost (127.0.0.1)
Host is up (0.000058s latency).
Other addresses for Localhost (not scanned): ::1
rDNS record for 127.0.0.1: localhost
Not shown: 997 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r--  1 0      0          0 Mar 26  2018 ftp_test
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 127.0.0.1
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 4
|   vsFTPD 3.0.3 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 8.2 (protocol 2.0)
23/tcp    open  telnet   Openwall GNU/*/Linux telnetd
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.31 seconds
[analyst@secOps ~]$
```

01 110 101 011 110 10 101 010

CyberOps Workstation [Running] - Oracle VirtualBox

Applications: [Mozilla Firefox] [Terminal - analyst@sec0...] Tue 28 Oct, 11:35 analyst

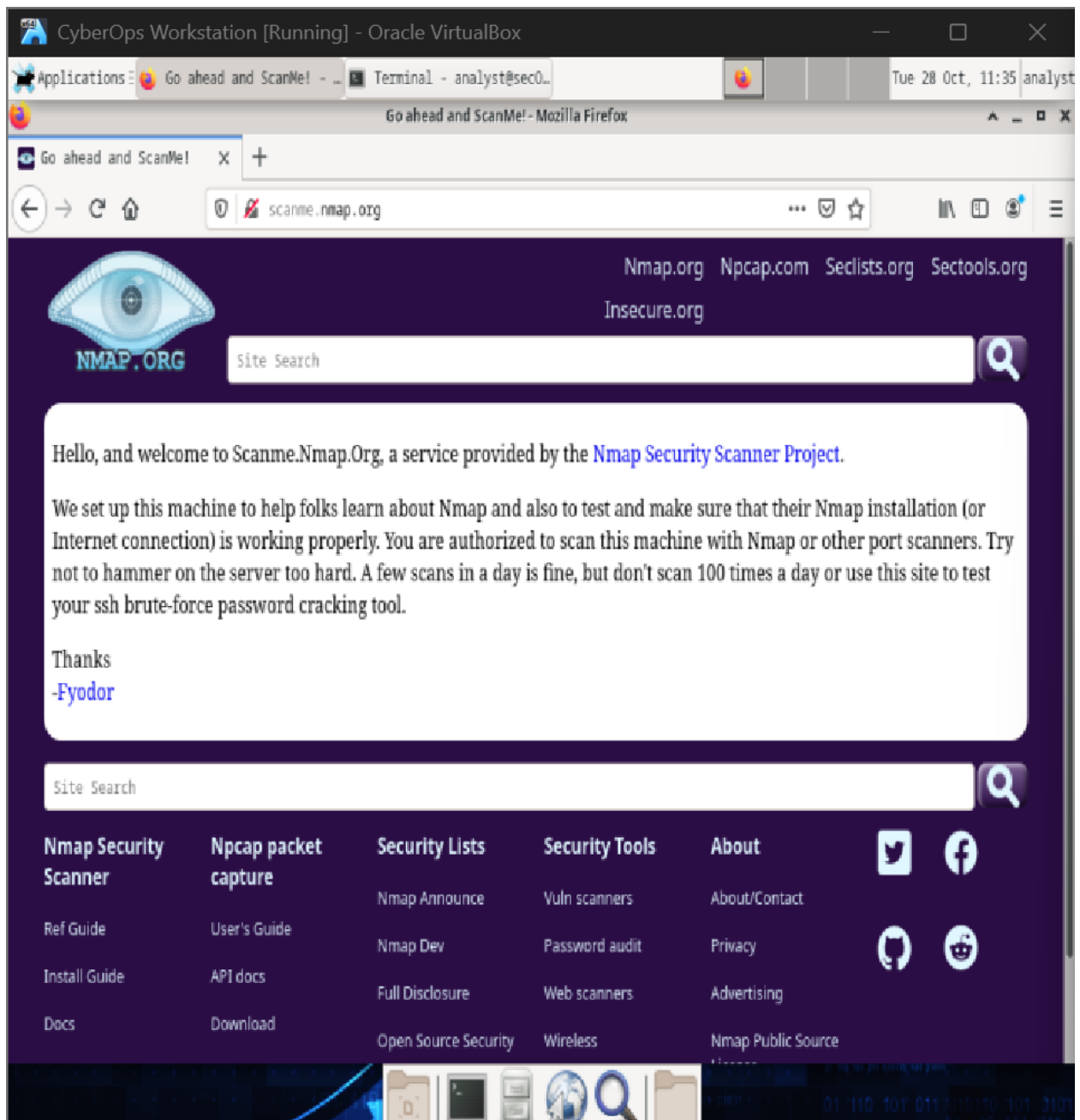
Terminal - analyst@secOps:-

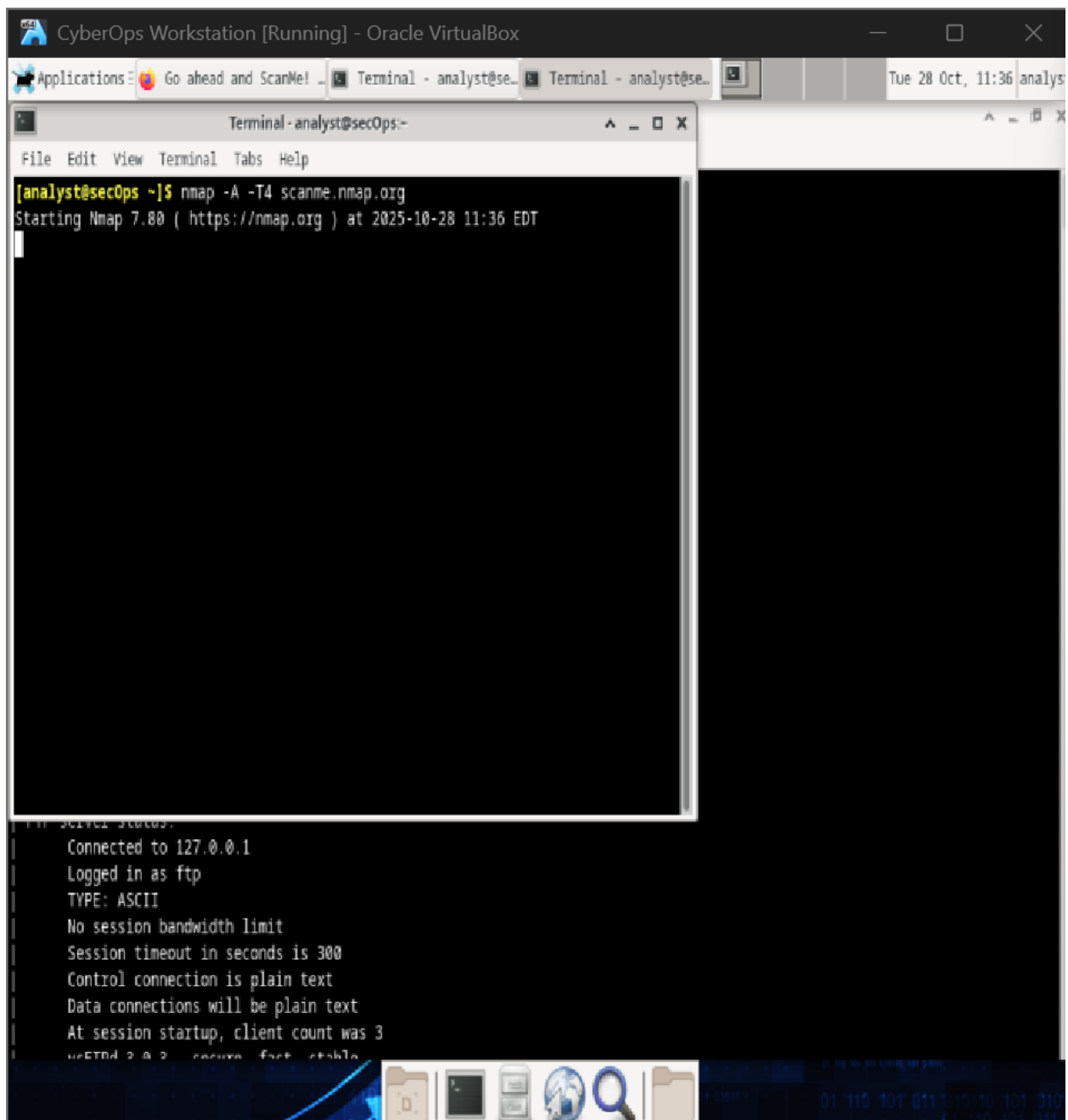
File Edit View Terminal Tabs Help

```
| No session bandwidth limit
| Session timeout in seconds is 300
| Control connection is plain text
| Data connections will be plain text
| At session startup, client count was 4
| vsFTPD 3.0.3 - secure, fast, stable
|_End of status
22/tcp open  ssh      OpenSSH 8.2 (protocol 2.0)
23/tcp open  telnet   Openwall GNU/Linux telnetd
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.31 seconds
[analyst@secOps ~]$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 16:4a:96:c2:cb:f3 brd ff:ff:ff:ff:ff:ff
3: s1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether f6:11:f0:e5:44:43 brd ff:ff:ff:ff:ff:ff
4: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7e:d0:17 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.5/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 80926sec preferred_lft 80926sec
    inet6 fe80::a00:27ff:fe7e:d017/64 scope link
        valid_lft forever preferred_lft forever
[analyst@secOps ~]$ nmap -A -T4 127.0.0.1/8
Starting Nmap 7.80 ( https://nmap.org ) at 2025-10-28 11:34 EDT
```

01 110 101 011 10110 101 0101





The screenshot shows a terminal window titled "Terminal - analyst@secOps:-" within a "CyberOps Workstation [Running] - Oracle VirtualBox" environment. The terminal displays the output of an Nmap scan command: `nmap -A -T4 scanme.nmap.org`. The output includes the Nmap version (7.80), the target IP (45.33.32.156), and a list of open ports with their corresponding services and versions. The scan also identifies the operating system as Linux.

```
[analyst@secOps ~]$ nmap -A -T4 scanme.nmap.org
Starting Nmap 7.80 ( https://nmap.org ) at 2025-10-28 11:36 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.20s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 994 closed ports
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|_ 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|_ 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Go ahead and ScanMe!
2525/tcp  filtered  ms-v-worlds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

data connections will be plain text
At session startup, client count was 4
vsFTPD 3.0.3 - secure, fast, stable
_End of status
22/tcp    open      ssh          OpenSSH 8.2 (protocol 2.0)
23/tcp    open      telnet       Openwall GNU/*/Linux telnetd
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Conclusions

This lab successfully demonstrated the capabilities of Nmap as a network scanning and security auditing tool. Key takeaways include:

1. **Technical Skills:** Gained hands-on experience with Nmap commands and switches

2. **Security Awareness:** Understanding both defensive and offensive uses of scanning tools
3. **Network Analysis:** Ability to interpret scan results and identify security concerns
4. **Ethical Understanding:** Recognition of the importance of authorization and responsible use

The scans revealed several security concerns, particularly:

- Anonymous FTP access on the localhost
- Telnet service running (unencrypted protocol)
- Multiple open services that could be attack vectors if not properly secured

Recommendations:

- Disable unnecessary services (especially Telnet)
- Implement strong authentication for FTP or disable anonymous access
- Regularly audit network services using tools like Nmap
- Keep all services updated to the latest secure versions
- Implement network segmentation and firewall rules